

Aditi Thanekar

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Education

University of California - San Diego

MS in Computer Science

La Jolla, California

September 2025 - December 2026

University of California - Riverside

BS in Computer Science, GPA: 3.76 / 4.0

Riverside, California

September 2022 - June 2025

Experience

Advanced Micro Devices (AMD)

San Jose, California

Compiler Engineering Intern - Triton | Triton/Gluon, AMD GCN, rocprofv3, LLVM

June 2026 - September 2026

- Optimized DeepSeek-v4 sparse attention kernel on MI350 GPUs with Triton/Gluon, achieving **75%** of roofline peak perf
- Integrated the optimized kernel into TokenSpeed serving engine and merged the contribution upstream, increasing end-to-end model throughput **12%** and reducing time to first token (TTFT) by **36%** on long context workloads
- Investigated LLVM scheduling constraints that limited instruction interleaving due to predicted spills; prototyped an additional post-register-allocation scheduling strategy to improve interleaving while avoiding spills
- Profiled rocprofv3 thread traces and analyzed AMD GCN assembly to identify bottlenecks and guide kernel optimization

Lawrence Livermore National Laboratory

Livermore, California

Computing Intern - National Ignition Facility | C++, CUDA, Linux, RAJA, GDB

June 2024 - November 2025

- **Summer 2025:** Fixed GPU execution bugs in RAJA-enabled laser simulation codebase and optimized FFTs
 - * Achieved **3.1x speedup** on GPU by integrating CuFFTW; migrated to Unified Memory, profiled w/Nsight Systems
 - * Refactored **3000+ lines** of C++ code to resolve CPU-GPU pointer issues in GPU loops, enabling stable execution
- **Summer 2024:** Developed a new C++ feature to group laser simulation component chains into reusable units
 - * Reduced Qt cold start time by **30%** and enabled reuse of component chains to cut pipeline creation time by **60%**
 - * Incorporated into the March 2025 **production release**.
- Authored LaTeX documentation, led live training for ~20 users, and presented work at poster symposium

Systems Optimization + Computer Architecture Lab

Riverside, California

Undergraduate Researcher - GPU Computing | Advisor: Daniel Wong

January 2025 - June 2025

- Ported MiniFE, an open-source finite element analysis code, from CUDA to AMD HIP for compatibility across GPUs
- Debugged race conditions and segmentation faults using cuda-gdb and rocgdb to trace memory errors
- Analyzed source code to identify warp divergence and memory coalescing inefficiencies, validated with nvprof and rocprof

University of California, Riverside

Riverside, California

Chief Grader | Data Structures and Algorithms - Professor Patrick Miller

April 2023 - June 2025

- Managed gradebook for **1000+** students, and directed tasks amongst 12 people, ensuring timely release of grades
- Conducted **200+ C++ code reviews** and live demos, assessing readability, efficiency, memory usage and best practices

Projects

3D Gaussian Splatting Renderer | Python, PyTorch

- Implemented end-to-end 3D Gaussian Splatting pipeline from scratch with self-captured multi-view dataset using COLMAP SfM point cloud for 3D reconstruction and novel view synthesis
- Built differentiable renderer with covariance projection and depth-sorted alpha blending for Gaussian splatting
- Improved reconstruction quality via photometric loss with densification, pruning, and background masking

FPGA Acceleration with HLS | C++, Vitis HLS, PYNQ-Z2

- Implemented hardware-accelerated kernels using High-Level Synthesis (HLS)
- Optimized latency and throughput via loop unrolling, pipelining, and memory partitioning
- Deployed designs on PYNQ-Z2 FPGA and analyzed performance vs CPU implementations

Technical Skills

Languages: C++, Python, Java

GPU/Accelerated Computing: CUDA, HIP, Triton/Gluon, AMD GCN

Systems & Performance: GDB, Git, GTest, rocprofv3, Nsight Systems, Rocprof Compute Viewer

Relevant Coursework: GPU Programming, Parallel Programming, Computer Architecture, Computer Graphics, AI